

Linear Relations

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math).

1. **[L1 · First differences]** Differences: $10 - 6 = 4$, $14 - 10 = 4$, $18 - 14 = 4$.
Constant (4), so the relation is linear.
2. **[L1 · Direct or partial]** y-intercept is 1, which is not 0.
So it is partial variation.
3. **[L1 · Slope]** slope = $(13 - 4)/(5 - 2) = 9/3$.
= 3.
4. **[L1 · y-intercept]** The constant term is the y-intercept: -8, at (0, -8).
5. **[L1 · x-intercept]** Let $y = 0$: $0 = 2x - 10 \rightarrow 2x = 10 \rightarrow x = 5$, at (5, 0).
6. **[L1 · Standard form]** Move all terms to one side: $4x - y + 1 = 0$.
7. **[L2 · First differences]** Differences: 3, 5, 7.
Not constant, so the relation is non-linear.
8. **[L2 · Direct variation]** $k = y/x = 15/5 = 3$.
Equation: $y = 3x$.
9. **[L2 · Slope]** slope = $(-4 - 8)/(1 - (-3)) = -12/4$.
= -3.
10. **[L2 · Both intercepts]** x-int: $y = 0 \rightarrow 3x = 24 \rightarrow (8, 0)$.
y-int: $x = 0 \rightarrow 4y = 24 \rightarrow (0, 6)$.
11. **[L2 · Equation of a line]** $5 = 2(3) + b \rightarrow 5 = 6 + b \rightarrow b = -1$.
 $y = 2x - 1$.
12. **[L2 · Standard form]** Multiply every term by 3: $3y = 2x - 12$.
Rearrange: $2x - 3y - 12 = 0$.
13. **[L3 · Missing table value]** The relation is linear, so first differences are equal.
From $y = 1$ to $y = 13$ over two steps is +12, so +6 each step.
Missing value: $1 + 6 = 7$.
14. **[L3 · Partial variation model]** $C = 15(6) + 20 = 90 + 20 = \110 .
The fixed value (joining fee) is \$20.
15. **[L3 · Unknown coordinate]** $(17 - 5)/(k - 2) = 4 \rightarrow 12 = 4(k - 2)$.
 $k - 2 = 3 \rightarrow k = 5$.

- 16. [L3 · Equation from intercepts]** Through $(-4, 0)$ and $(0, 8)$: slope = $(8 - 0)/(0 - (-4)) = 8/4 = 2$.
y-intercept is 8, so $y = 2x + 8$.
- 17. [L3 · Line through two points]** slope = $(14 - 2)/(3 - (-1)) = 12/4 = 3$.
 $2 = 3(-1) + b \rightarrow 2 = -3 + b \rightarrow b = 5$.
 $y = 3x + 5$.
- 18. [L3 · Correlation reasoning]** A strong positive linear correlation.
As temperature rises, ice-cream sales tend to rise, so temperature is a good predictor of sales.